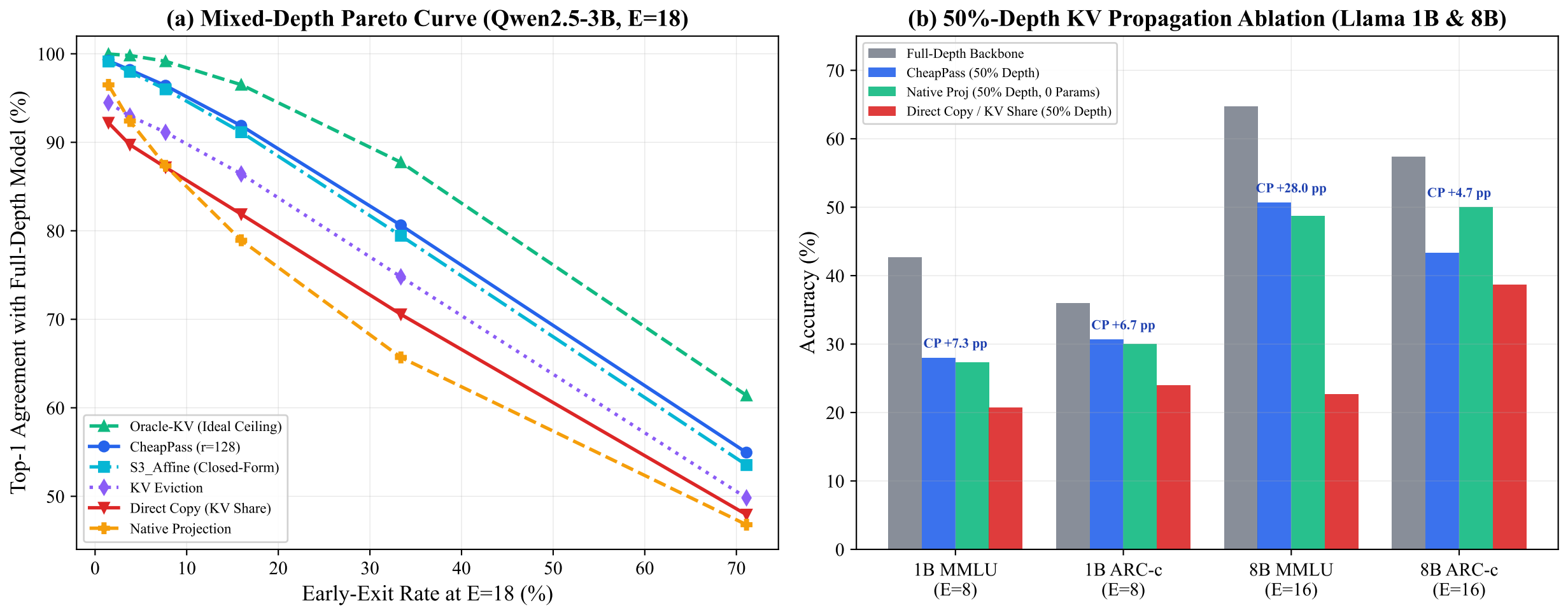
(a) Mixed-Depth Pareto Curve (Qwen2.5-3B, E=18)
Top-1 Agreement with Full-Depth Model (%)
Early-Exit Rate at E=18 (%)
Oracle-KV (Ideal Ceiling)
CheapPass (r=128)
S3_Affine (Closed-Form)
KV Eviction
Direct Copy (KV Share)
Native Projection
(b) 50%-Depth KV Propagation Ablation (Llama 1B & 8B)
Accuracy (%)
Full-Depth Backbone
CheapPass (50% Depth)
Native Proj (50% Depth, 0 Params)
Direct Copy / KV Share (50% Depth)
CP +7.3 pp
CP +6.7 pp
CP +28.0 pp
CP +4.7 pp
1B MMLU (E=8)
1B ARC-c (E=8)
8B MMLU (E=16)
8B ARC-c (E=16)